

BOULDER 14 W, CO, USA

WMO: 745200

Lat: 40.0354N

Lon: 105.5409W

Elev: 9828

StdP: 10.17

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 94075

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-6.2	-1.2	-17.3	3.1	7.2	-13.1	3.9	10.0	25.0	20.5	23.8	22.1	4.2	N/A	0.687

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	71.4	48.1	69.2	48.0	67.0	47.7	53.5	61.2	52.2	61.2	50.9	60.9	4.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.9	80.3	56.0	49.1	75.0	54.9	47.2	69.7	53.7	26.6	61.4	25.5	61.3	24.6	60.7	59.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.1	19.7	17.2	DB	-17.5	75.9	6.7	1.9	-22.3	77.3	-26.2	78.4	-29.9	79.5	-34.8	80.9	(4)
(5)			WB	-18.8	57.1	7.8	1.2	-24.4	57.9	-28.9	58.6	-33.3	59.3	-38.9	60.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.6	20.6	19.5	26.5	31.2	38.8	49.7	55.0	52.8	47.2	36.6	27.8	19.9		
	DBStd	15.02	9.41	10.06	9.28	8.31	7.52	5.89	4.26	4.31	6.42	8.89	9.87	10.33		
	HDD50	5658	913	855	730	565	351	75	10	23	121	418	665	932		
	HDD65	10749	1378	1275	1195	1015	814	458	311	377	533	881	1115	1397		
	CDD50	385	0	0	0	0	2	68	165	110	38	2	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	11	0	0	0	0	0	4	6	1	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	6.0	8.9	8.2	6.9	5.7	4.4	4.3	3.8	3.8	4.2	5.9	7.3	8.4		
	PrecAvg	18.5	1.2	1.1	1.5	2.0	2.0	1.5	2.0	2.0	1.7	1.2	1.2	1.1		
(15) Precipitation	PrecMax	28.9	3.1	2.4	4.5	4.7	5.0	3.3	4.5	3.8	12.4	2.2	2.0	2.5		
	PrecMin	13.1	0.4	0.3	0.1	0.2	0.7	0.3	0.0	0.7	0.0	0.3	0.5	0.2		
	PrecStd	3.4	0.6	0.6	0.9	1.1	1.0	0.8	1.1	0.8	2.2	0.6	0.4	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.7	44.6	53.3	56.5	65.0	74.1	74.7	71.4	70.2	61.4	51.8	43.7		
		MCWB	29.9	30.5	35.3	37.7	42.0	47.9	49.0	48.2	46.1	41.0	34.8	28.7		
	2%	DB	39.9	40.6	48.5	52.6	61.1	70.6	71.9	69.0	66.3	57.7	47.8	39.3		
		MCWB	28.0	28.7	33.4	36.5	41.4	46.8	48.9	48.1	46.0	39.7	33.0	27.0		
	5%	DB	36.8	36.7	44.5	49.1	57.4	67.8	69.8	67.1	63.6	54.6	44.9	36.3		
		MCWB	26.4	26.8	31.1	34.4	40.0	46.1	49.3	47.8	44.2	38.6	31.7	26.0		
	10%	DB	33.7	33.3	40.8	45.1	53.5	64.6	67.5	64.8	60.9	51.0	41.6	33.7		
		MCWB	25.2	25.0	29.5	32.8	39.1	45.3	49.1	47.7	43.9	37.2	29.7	25.1		
	0.4%	WB	31.7	32.3	36.5	39.7	45.7	54.1	55.9	54.3	52.7	44.6	36.3	30.5		
		MCDB	41.7	42.0	51.1	52.5	55.5	61.5	62.9	61.2	61.2	54.3	49.4	39.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	29.3	29.8	34.1	37.3	42.9	50.9	54.2	52.5	49.8	42.1	33.9	29.0		
		MCDB	36.8	39.7	47.4	50.2	53.2	61.2	61.9	61.2	58.4	52.2	45.7	36.7		
	5%	WB	27.7	27.7	32.1	35.4	41.3	49.4	53.0	51.1	47.7	40.2	32.1	27.6		
		MCDB	35.6	36.2	43.3	46.4	52.9	61.7	61.3	61.4	57.9	51.3	42.9	35.1		
	10%	WB	25.9	26.2	30.2	33.5	39.6	47.8	51.6	49.9	46.2	38.0	30.5	26.1		
		MCDB	33.3	33.6	39.9	43.2	51.8	60.8	61.1	60.6	57.2	49.3	40.2	33.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	15.1	17.0	17.0	19.1	24.1	22.7	22.7	22.3	18.2	15.5	14.5		
		MCDBR	17.6	16.5	20.5	23.3	26.9	28.7	26.7	25.7	26.3	24.2	18.4	15.8		
		MCWBR	10.7	9.5	10.1	11.3	12.5	12.4	11.6	11.5	11.5	11.3	10.0	9.4		
		MCDBR	16.3	14.2	19.3	21.5	23.8	25.2	22.0	22.3	21.7	21.2	18.4	15.2		
	5% WB	MCWBR	10.6	9.0	10.0	11.5	12.8	13.6	12.3	11.7	11.3	11.5	9.9	9.9		
(40) Clear-Sky Solar Irradiance	taub		0.206	0.212	0.230	0.256	0.270	0.276	0.298	0.295	0.259	0.238	0.219	0.210		
	taud		2.611	2.581	2.577	2.567	2.607	2.605	2.571	2.563	2.686	2.712	2.680	2.626		
	Ebn at Noon		321	331	331	325	320	316	308	308	316	317	314	312		
	Edh at Noon		21	25	28	30	30	30	31	30	25	22	19	19		
(44) All-Sky Solar Radiation	RadAvg		834	1164	1540	1750	1889	2200	1996	1773	1596	1185	846	705		
	RadStd		38	63	129	119	154	125	96	89	103	79	55	45		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page